Invariant subspaces after complexification of a real irreducible operator

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Abstract

We construct a real irreducible bounded operator whose complexification has exactly two proper nonzero closed invariant subspaces. A real polynomial approximation recovers the complex structure, and spectral separation determines the complexified invariant-subspace lattice. We compare the construction with the original conjecture and a later proposition.

Repository target. This manuscript concerns AIM problem 920 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

The catalogue asks whether a bounded operator on an infinite-dimensional separable real Banach space, having no proper nonzero closed real invariant subspace, must retain that property after complexification. This is Conjecture 3 in [1]. The following construction gives a negative answer. It uses the classical existence of an operator on a complex Banach space with no proper nonzero closed complex invariant subspace; the passage to the required real operator is elementary.

Theorem 1.1. *There are an infinite-dimensional separable real Banach space X and a bounded real-linear operator T such that*

$$\mathrm{Lat}_{\mathbb{R}}(T) = \{\{0\}, X\}, \qquad \mathrm{Lat}_{\mathbb{C}}(T_{\mathbb{C}}) = \{\{0\}, E_+, E_-, X_{\mathbb{C}}\},$$

where E_+, E_- are nonzero closed complementary complex subspaces. One may take X to be the real space underlying $\ell_1(\mathbb{C})$, or equivalently transfer the example to $\ell_1(\mathbb{R})$. The operator can also be chosen invertible and of norm one.

Lemma 1.2. *Let Y be a complex Banach space, let $X = Y_{\mathbb{R}}$ be its underlying real space, and let $Jx = ix$ denote its original complex structure. If $S \in \mathcal{L}_{\mathbb{C}}(Y)$ and $\|S\| \le 1/10$, then J lies in the norm-closed unital real algebra generated by $T = J + S$.*

Proof. We have $J^2 = -I$, $\|J\| = 1$, and $JS = SJ$. Define

$$Q = I - JS, \qquad E = -T^2 - I = -2JS - S^2.$$

Then $T = JQ$, $Q^2 = I + E = -T^2$, $\|Q - I\| \le 1/10$, and $\|E\| \le 21/100 < 1$. In the norm-closed real algebra generated by T, the real binomial series

$$U = \sum_{n=0}^{\infty} \binom{1/2}{n} E^n$$

converges absolutely and satisfies $U^2 = I + E = Q^2$. Moreover

$$\|U - I\| \le \sum_{n\ge 1} \left| \binom{1/2}{n} \right| (21/100)^n = 1 - \sqrt{79/100} < \frac{12}{100}.$$

The operators U, Q commute, because Q commutes with E. Also $\|(U + Q) - 2I\| < 22/100 < 2$, so $U + Q$ is invertible. Factoring $U^2 - Q^2$ yields $U = Q$. Since $\|U - I\| < 1$, the inverse U^{-1} is a Neumann series in the same closed real algebra. Therefore $J = TU^{-1}$ belongs to that algebra, as asserted. □

Proof of Theorem 1.1. Choose a bounded complex-linear operator R on $Y = \ell_1(\mathbb{C})$ with no proper nonzero closed complex invariant subspace. This is the classical Read construction [2]. For an explicit modern primary source covering the complex field, the introduction of [3] allows real or complex spaces, and its Theorem 1.1 gives the stronger absence of nontrivial invariant closed subsets on $\ell_1(\mathbb{C})$: take $Z = \ell_1(\mathbb{C})$ and use the isometry $\ell_1(Z) \cong \ell_1(\mathbb{C})$.

Multiply R by a nonzero real scalar to obtain S with $\|S\| \le 1/10$. Its complex invariant subspaces are unchanged. Regard Y as the real space X and set $T = J + S$. If a closed real subspace M is T-invariant, it is invariant under every real polynomial in T, and therefore under their operator-norm limits. Lemma 1.2 gives $J(M) \subseteq M$. Thus M is a complex subspace of Y, and $S(M) = (T - J)(M) \subseteq M$. By the choice of S, either $M = \{0\}$ or $M = X$. This proves real irreducibility.

Now form the external complexification $X_{\mathbb{C}} = X + iX$, with the Taylor norm specified in the catalogue. The external scalar i is distinct from the original complex structure J on X. The bounded complex-linear extension $J_{\mathbb{C}}$ commutes with $T_{\mathbb{C}}$ and satisfies $J_{\mathbb{C}}^2 = -I$. Hence

$$P_+ = \frac{I - iJ_{\mathbb{C}}}{2}, \qquad P_- = \frac{I + iJ_{\mathbb{C}}}{2}$$

are complementary bounded projections commuting with $T_{\mathbb{C}}$. Their ranges are

$$E_+ = \ker(J_{\mathbb{C}} - iI) = \{x - iJx : x \in X\}, \qquad E_- = \ker(J_{\mathbb{C}} + iI) = \{x + iJx : x \in X\}.$$

For every $x \neq 0$ both displayed vectors are nonzero. Thus these are proper nonzero closed complex invariant subspaces, and $X_{\mathbb{C}} = E_+ \oplus E_-$.

We also identify the entire invariant-subspace lattice. Complexification is norm-continuous for the Taylor norm; indeed $\|A_{\mathbb{C}}\| = \|A\|$ for every bounded real operator A. Lemma 1.2 therefore implies that $J_{\mathbb{C}}$ is a norm limit of real polynomials in $T_{\mathbb{C}}$. Every closed complex $T_{\mathbb{C}}$-invariant subspace N is consequently invariant under P_+, P_-, and

$$N = (N \cap E_+) \oplus (N \cap E_-).$$

The map $x \mapsto x - iJx$ is a complex-linear topological isomorphism from Y to E_+, intertwining $S + iI$ with $T_{\mathbb{C}}|_{E_+}$. The analogous map to E_- is conjugate-linear, so its restriction is likewise complex irreducible. Each intersection is therefore either zero or the whole corresponding $E_\pm$, proving the asserted four-element lattice.

Finally $T = JQ$ is invertible because $\|Q - I\| \le 1/10$. Multiplying T by $\|T\|^{-1}$ changes neither invariant-subspace lattice. The real-linear identification

$$(a_1, b_1, a_2, b_2, \ldots) \longmapsto (a_1 + ib_1, a_2 + ib_2, \ldots)$$

is an isomorphism $\ell_1(\mathbb{R}) \to Y_{\mathbb{R}}$, since

$$\sum_n \sqrt{a_n^2 + b_n^2} \le \sum_n (|a_n| + |b_n|) \le \sqrt{2} \sum_n \sqrt{a_n^2 + b_n^2}.$$

Transporting T through this isomorphism gives the claimed example on the classical real ℓ_1 space. □

An independent consequence of an existing theorem

There is also a shorter route using the stronger result already proved in [3, Theorem 1.1]. On $Y = \ell_1(\mathbb{C})$ it supplies an operator R for which every nonzero vector has dense forward orbit. The topology is unchanged on passing to the underlying real space. Therefore every nonzero closed real R-invariant subspace contains a dense orbit and equals $Y_{\mathbb{R}}$. Nevertheless the external complexification of R preserves the nonzero proper eigenspaces of the original complex structure $J_{\mathbb{C}}$, exactly as above. Thus a negative answer to the stated catalogue problem is also a direct consequence of that existing theorem. This observation is not a claim that the existence theorem or this consequence is new. The preceding proof needs only the weaker classical invariant-subspace counterexample and additionally identifies the complete complexified invariant-subspace lattice.

Conflict with a cited proposition

There is a specific literature conflict that should accompany this answer. Proposition 4.1, printed p. 184 of [4], asserts equivalence of real and complexified invariant-subspace existence for real Banach lattices whose order is induced by an unconditional basis. It imposes no positivity or band-diagonal assumption on the operator. The global convention in its Section 1.1 restricts the spaces to such lattices; it defines $\mathcal{L}(X)$ as all bounded linear operators and introduces positivity separately. The surrounding Section 4 does not impose an additional standing positivity assumption. The paragraph preceding it cites Chapter 9 of the book by Chalendar and Partington as general background on moment sequences; no proof of the proposition is supplied there.

The construction above also satisfies that space hypothesis. On $Y_{\mathbb{R}}$ the ordered real basis $(e_1, ie_1, e_2, ie_2, \ldots)$ is 1-unconditional and its coordinate order is a Banach lattice order: the norm is $\sum_n \sqrt{a_n^2 + b_n^2}$. Alternatively use the transported example on $\ell_1(\mathbb{R})$. It therefore contradicts implication (i)⇒(ii) of that proposition as printed. The elementary opposite implication, obtained by complexifying a real invariant subspace, remains valid. Our example is not a positive band-diagonal operator, so it does not by itself contradict the positive band-diagonal existence theorems elsewhere in that paper.

The original formulation [1, Conjecture 3] has also been checked with its preceding assumptions: it concerns arbitrary bounded operators on infinite-dimensional separable real spaces, with no additional spectral restriction. Its Lemma 5 describes a commuting closed complex structure as the obstruction to the conjecture; here that structure is the bounded, everywhere-defined J recovered above. The proof provides the counterexample independently of the conflicting proposition. Priority and the reason for the published discrepancy have not been established in this investigation.

References

[1] Y. A. Abramovich, C. D. Aliprantis, G. Sirotkin, and V. G. Troitsky, *Some open problems and conjectures associated with the invariant subspace problem*, Positivity **9** (2005), 273–286. doi:10.1007/s11117-004-3786-9. Conjecture 3.

[2] C. J. Read, *A short proof concerning the invariant subspace problem*, J. London Math. Soc. (2) **34** (1986), 335–348. doi:10.1112/jlms/s2-34.2.335.

[3] S. Grivaux and M. Roginskaya, *A general approach to Read's type constructions of operators without non-trivial invariant closed subspaces*, Proc. London Math. Soc. (3) **109** (2014), 596–652. doi:10.1112/plms/pdu012. arXiv:1301.6143, Theorem 1.1.

[4] E. A. Gallardo-Gutiérrez and F. J. González-Doña, *Band-diagonal operators on Banach lattices: Matrix dynamics and invariant subspaces*, Linear Algebra Appl. **674** (2023), 155–191. doi:10.1016/j.laa.2023.05.017. The conflict discussed in Section 1 concerns Proposition 4.1.